INCH -POUND

MIL-DTL-17505H
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SUPERSEDING
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DETAIL SPECIFICATION

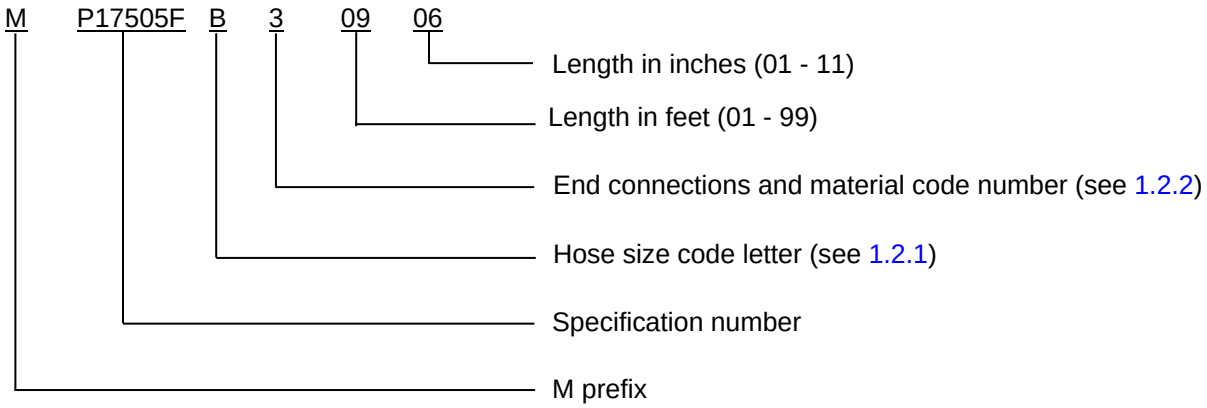
HOSE AND HOSE ASSEMBLY, RUBBER, OIL AND GASOLINE, SUCTION AND DISCHARGE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers hose and hose assembly, rubber oil and gasoline, suction and discharge. The hose and hose assemblies for suction and discharge service in the transfer of fuels, such as Navy special fuel oil, diesel fuel, oil, jet fuel (JP-5 or JP-8), motor gasoline, and aviation gasoline. The hose is intended to handle petroleum products having an aromatic content no greater than 30 percent.

1.2 Part or Identifying Number (PIN). The PIN consists of the letter M, the letter P, specification number, a letter for hose size, a number for end connections, two numbers for length in feet, and two numbers for length in inches.



PIN examples:

MP17505FB30906 describes a hose assembly 6 inch ID, bronze flange and nipple, 9 feet 6 inches long.
MP17505B-2500 describes a hose 6 inch ID, 25 feet long.

1.2.1 Hose size. Hose size is designated by one code letter (see table I).

Comments, suggestions, or questions on this document should be addressed to: DLA Land and Maritime, Attn: VAI, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to FluidFlow@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A

FSC 4720

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TABLE I. Hose size code letter.

Hose size code letter	Hose ID inches (mm)
A	4 (102 mm)
B	6 (152 mm)
C	8 (203 mm)
D	12 (305 mm)

1.2.2 End connections material. Material of hose is designated by a one-digit code number (see table II).

TABLE II. Hose end connections material code numbers.

End connection material code	End connections	Material		
		Nipple	Flange	Swivel coupling
- (dash)	None	---	---	---
1	Flanges and nipples	Steel	Steel	---
2	Flanges and nipples	Aluminum	Aluminum	---
3	Flanges and nipples	Bronze	Bronze	---
4	Flanges and nipples	Brass	Brass	---
5	Swivel couplings and nipples			Copper 82.0 - 84.0% Tin 2.5 - 3.5% Lead 2.5 - 3.5% Iron Max - .035 Nickel Max - .50 Aluminum ---

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 or 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 or 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

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FEDERAL STANDARDS

FED-STD-H28	-	Screw-Thread Standards for Federal Services
FED-STD-162	-	Hose, Rubber, Visual Inspection Guide for
FED-STD-601	-	Rubber, Sampling, and Testing

(Copies of these documents are available online at http://quicksearch.dla.mil/.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the cited in the solicitation or contract.

ASME INTERNATIONAL

ASME B16.5	-	Pipe Flanges and Flanged Fittings NPS 1/2 Through NPS 24 Metric/Inch Standard

(Copies of these documents are available online at http://www.asme.org.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B16.24	-	Cast Copper Alloy Pipe Flanges and Flanged Fittings, Class 150, 300, 400, 600, 900, 1,500 and 2,500

(Copies of these documents are available on-line at http:/www.ansi.org.)

ASTM INTERNATIONAL

ASTM B26/B26M	-	Standard Specification for Aluminum-Alloy Sand Castings.
ASTM B584	-	Standard Specification for Copper Alloy Sand Castings for General Applications.
ASTM D297	-	Standard Test Methods for Rubber Products-Chemical Analysis.
ASTM D380	-	Standard Test Methods for Rubber Hose.
ASTM D381	-	Standard Test Method for Gum Content in Fuels by Jet Evaporation.
ASTM D412	-	Standard Test Methods for Vulcanized Rubber and Thermoplastic Elastomers-Tension.
ASTM D413	-	Standard Test Methods for Rubber Property-Adhesion to Flexible Substrate.
ASTM D471	-	Standard Test Method for Rubber Property-Effect of Liquids.
ASTM D573	-	Standard Test Method for Rubber-Deterioration in an Air Oven.
ASTM D1149	-	Standard Test Methods for Rubber Deterioration-Cracking in an Ozone Controlled Environment.
ASTM D2702	-	Standard Practice for Rubber Chemicals-Determination of Infrared Absorption Characteristics.
ASTM E8/E8M	-	Standard Test Methods for Tension Testing of Metallic Materials.

(Copies of these documents are available online at http://www.astm.org.)

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INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO/IEC DIS 17025 - General Requirements for the Competence of Testing and Calibration Laboratories

(Copies of these documents are available online at www.ansi.org.)

NCSL INTERNATIONAL

NCSL Z540.3 - Requirements for the Calibration of Measuring and Test Equipment

(Copies of these documents are available online at http://www.ncsli.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.3.

3.2 Materials. Materials used shall be free from defects, which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use.

3.2.1 Recycled, recovered, environmentally preferable or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.2.2 Recovered materials. The term “recovered materials” means materials, which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. Unless otherwise specified, none of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification.

3.2.3 Rubber. The basic synthetic material used for the tube shall be a copolymer product. The basic synthetic materials used for the cover shall be of an oil, abrasion, oxidation weathering, and sunlight resistant material.

3.2.4 Wire reinforcement. The reinforcing wire or wires shall be round steel. The wire shall have the applicable tensile strength shown in table III when tested in accordance with 4.7.18.1. The wire shall be of such area as to enable the hose to meet the kink test specified in 3.9.

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TABLE III. Physical properties of wire. 1/

Wire diameter, inch	Tensile strength, psi	
	Minimum	Maximum
0.035 to 0.105 (1 to 3 (mm)) inclusive	180,000 (1241 MPa)	300,000 (2 068 MPa)
More than 0.105 to 0.207 (3 to 5 mm) inclusive	150,000 (1035 MPa)	240,000 (1654 MPa)
More than 0.207 to 0.362 (5 to 9 mm) inclusive	120,000 (827 MPa)	210,000 (1448 MPa)

1/ Metric equivalents are given for information only.

3.2.4.1 Flexibility of wire. The wire shall show no defects in the steel when tested as described in 4.7.18.2.

3.2.5 Carcass. Textile reinforcement of the hose shall be of synthetic fiber cord or woven fabric.

3.3 Length. The length of hose shall be as specified (see 6.2), when measured in accordance with 4.7.1. The specified length shall be measured from face to face of the nipple, fitting, or coupling ends on hose.

3.3.1 Tolerances. For 20 feet (6096 mm) and less, a tolerance of ± 2.5 inches (63.5 mm), for hose lengths over 20 feet (6096 mm), a tolerance of ± 1 percent shall be permitted in the length of hose.

3.4 Construction. The hose is intended to handle petroleum products having an aromatic content no greater than 30 percent. The hose assembly shall be constructed as specified in 3.4.1 through 3.4.7 and in accordance with the dimensions and materials indicated in table IV. The construction specified herein represents minimum requirements and is not intended to exclude hose having additional layers of textile reinforcement or compound, or hose having special sections of wire that will not sacrifice smoothness of bore or resistance to damage. The maximum working pressure of this hose is 150 psi (1034 kPa).

TABLE IV. Physical requirements of hose. 1/ 2/

Requirements	4-inch (102 mm) hose	6-inch (152 mm) hose	8-inch (203 mm) hose	12-inch (305 mm) hose
ID tolerance inch (mm)	±0.062 (2 mm)	+0.062 (+2 mm) -0.250 (-6 mm)	+0.062 (+2 mm) -0.250 (-6 mm)	+0.062 (+2 mm) -0.250 (-6 mm)
Thickness of tube - inch (mm) (min)	0.110 (3 mm)	0.110 (3 mm)	0.125 (3 mm)	0.125 (3 mm)
Thickness of cover - inch (mm) (min)	0.093 (2 mm)	0.110 (3 mm)	0.110 (3 mm)	0.110 (3 mm)
Minimum bend radii under pressure (feet (mm)) 3/	3.5 (1067 mm)	5.25 (1600 mm)	7.0 (2134 mm)	10.75 (3277 mm)

1/ Dimensions are in inches
2/ Metric equivalents are given for information only.
3/ Does not apply to hose less than 25 feet (7620 mm) long in 12-inch (305 mm) size.

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3.4.1 Hose body. In addition to the description of materials specified in table IV, the body of the hose shall consist of the following:

a. A synthetic rubber inner tube, properly compounded to resist the following:
   (1) Navy special fuel oil
   (2) Diesel fuel
   (3) Petroleum oils
   (4) Jet fuel (JP-5 or JP-8)
   (5) Motor gasoline
   (6) Aviation gasoline
   (7) Salt water
b. Reinforcement shall consist of either braided or spiraled yarn or multiple plies of cord or woven fabric.
c. A helix of round wire or wires thoroughly embedded in synthetic rubber, as specified in 3.2.5, and connected to the hose nipple by a suitable wire to ensure static bond.
d. Reinforcement as described under b. shall be applied over the helix or wire.
e. An oil and abrasion-resistant synthetic rubber cover compounded in accordance with 3.2.3 to give maximum resistance to oil, abrasion, oxidation weathering, salt water, and sunlight aging.

3.4.2 Tube and cover. The tube shall be smooth and free from pitting. The tube and cover shall be of uniform thickness and free from foreign substances. Phosphate plasticizers shall not be used in this product. The gage shall be in accordance with table IV.

3.4.3 Rubber filling strip. The space between adjacent wires shall be filled with strips of a synthetic rubber compound.

3.4.4 Hose ends.

3.4.4.1 Hand-built wire reinforced construction. Hose ends on the hand-built wire reinforced hose shall be capped by application of the synthetic rubber compound used in the cover of the hose. The following details shall apply:

a. The end configuration may be square, using the cover compound for a cap, or the end may be tapered so the cover compound will continue to the hose nipple.
b. Each length of hose shall be fitted with either a nipple with flanges attached or swaged-on permanent-type couplings with flanges attached.
c. Unless otherwise specified (see 6.2), the 4-inch (102 mm) size shall have integral swivel couplings (male and female) attached.

3.4.4.2 Horizontal braided/spiraled wire reinforced construction. Hose ends on the horizontal braided/spiraled-wire reinforced hose shall be sealed using the same compound used in the cover of the hose. Each length of hose shall be fitted with swaged-on permanent-type couplings, and shall have flanges attached thereto (see 6.4).

3.4.5 Textile reinforcement. Cotton of synthetic cord or woven fabric shall be free form unsightly defects, dirt, knots, lumps, and irregularities of twist.

3.4.6 Nipples and flanges. For oil transfer, steel, bronze, or brass nipples and flanges, or integral flanged aluminum-alloy nipples may be used. The following details shall apply:

a. For gasoline JP-5 and JP-8 transfer, bronze, brass, or integral flanged aluminum-alloy nipples and flanges shall be used, as specified (see 6.2).

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b. Steel nipples shall be cut from standard pipe and the exterior ends shall be consistent with the type of flanges to be used with regards to fastening.
c. If the nipple is threaded, it shall be in accordance with FED-STD-H28.
d. Two bands shall be welded or shrunk not less than 2 inches (51 mm) apart on the exterior of the nipple.
e. The steel flanges shall be screwed, welded, or welded slip-on type of forged steel in the 150-pound (68 kilograms (kg)) class, with dimensions, facings, and drillings in accordance with ASME B16.5
f. Bronze and brass flanges shall be screwed, welded, or welded slip-on type in the 150-pound (68 kg) class with dimensions, facings, and drillings in accordance with ANSI B16.24.
g. Use of epoxy resin as a thread sealant is acceptable.

3.4.6.1 Integral flanged aluminum-alloy nipples. The composition of the integral flanged aluminum-alloy nipple shall conform to UNS alloy A04430 or ANSI alloy B443.0, temper F. The following details shall apply:

a. The nipple shall be reinforced to withstand usage comparable to that of the steel specified in, or alloy 356.0 temper T6 of ASTM B26/B26M.
b. The fillet of the flanged nipple shall be reinforced to withstand usage comparable to that of the steel specified in 3.4.6.
c. The nipple portion of the aluminum-alloy nipple shall have cast bands on the exterior of the nipple. Nipples shall be made an integral part of the hose by means of a round; steel wire spiraled over the textile carcass over the nipple area.
d. The helix wire or wires shall be carried over the nipple area to provide additional anchorage.
e. End blocks of plies of fabric impregnated with synthetic rubber compound shall be applied over the nipple area tapering into the hose body.
f. The integral swivel couplings shall be secured to the 4-inch (102 mm) hose in the same manner and shall protrude 12 inches (305 mm) into the hose.
g. As an option, assemblies may be assembled using aluminum-alloy pipe with aluminum bands welded on and forged aluminum-alloy slip-on welding flanges.
h The assemblies shall maintain strength and tightness.
i. The flange dimensions and drillings shall be the same as specified for steel flanges except that the flange shall not have a raised face.

3.4.7 Integral swivel connection couplings. The integral swivel couplings for the 4-inch (102 mm) hose shall have two bands cast, brazed, or shrunk into the exterior. Construction shall be such that the integral nipples shall interface with the hose in the same manner as do the nipples. The composition of the integral swivel couplings shall conform to ASTM B584, copper alloy, UNS No. C84400.

3.5 Resistance to vacuum. The hose shall not collapse or blister or show any other indication of failure when subjected to a vacuum as specified in 4.7.4.

3.6 Volume swell of tube and cover compound. The percentage of swelling of the tube and cover in the special testing fuel, as determined by the method specified in 4.7.11, shall be not greater than 50 percent for the tube and 80 percent for cover.

3.7 Soluble nonvolatile requirement of tube compound. When tested as described in 4.7.16, the soluble nonvolatile matter in the tube shall be not greater than 6 percent.

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3.8 Hydrostatic pressure requirements.

3.8.1 Hydrostatic proof pressure. Each hose assembly shall be tested described in 4.7.5 and shall withstand an internal pressure of 250 psi (1724 kPa) without leakage or indication of weakness or deterioration.

3.8.2 Stability (dimensional changes). When subjected to the 250 psi (1724 kPa) hydrostatic pressure test of 4.7.5, the hose shall not change in length more than 7.5 percent of the original length. The permanent elongation shall be not greater than 2.5 percent.

3.8.3 Minimum bend test. A length of hose, when tested as specified in 4.7.6, shall not be damaged.

3.8.4 Burst pressure. The hose shall have a burst pressure of not less than 600 psi (4137 kPa). The burst test shall be conducted in accordance with 4.7.7. The hose assembly shall not leak at fittings or show any signs of slippage of the couplings before the hose fails.

3.9 Kink test. A length of hose shall be subjected to the test described in 4.7.8 and shall not kink or flatten under this test at no internal pressure.

3.10 Electrical conductivity (see 4.7.19). A low-resistance connection shall be provided by bonding the nipples to the built-in hose reinforcing wire or wires. The wire shall be so applied that the bonding will not be broken by flexing of the hose in service, and will ensure elimination of static at all times. The resistance shall not be greater than 100 ohms per foot.

3.11 Physical properties of tube and cover.

3.11.1 Tube and cover before aging or immersion. Prior to subjection to the immersion and aging tests specified in 4.7.10 and 4.7.12, the tensile strength of the tube shall be not less than 1,250 psi (8618 kPa). The tensile strength of the cover shall be 1,800 psi (12410 kPa) when tested as specified in 4.7.9. The ultimate elongation of the tube and cover of the hose shall be not less than 250 percent for the tube and 300 percent for the cover when tested in accordance with 4.7.9.

3.11.2 Tube and cover after immersion. After subjection to the immersion test specified in 4.7.10, the tensile strength of the tube and cover shall be not less than 40 percent of the original tensile strength or 600 psi (4137 kPa) (whichever is greater). Elongation shall be not less than 150 percent.

3.11.3 Tube and cover after accelerated aging of compounds. After aging as specified in 4.7.12, the tensile strength of the tube and cover shall be not less than 80 percent of the original tensile strength. After aging as specified in 4.7.12, the ultimate elongation of the tube and cover shall be not less than 50 percent of the original ultimate elongation.

3.12 Adhesion. The adhesion between the component parts of the hose shall conform to the requirements listed in table V, when tested as specified in the referenced paragraphs.

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TABLE V. Minimum loads for adhesion test. 1/

Adhesion Pounds-force per inch (lb-f/in) (kPa) minimum	Tube	Cover	Reinforcement plies	Test paragraphs
Initial	23 (159 kPa)	23 (159 kPa)	23 (159 kPa)	4.7.14
After filling	17 (117 kPa)	17 (117 kPa)	17 (117 kPa)	4.7.15

1/ Metric equivalents are given for information only.

3.13 Ozone resistance of hose cover. The cover stock of the hose at the end of the exposure time (see 4.7.13) shall show no visible cracking under 7X magnification.

3.14 Marking. Each length of hose shall be branded with manufacturer's name and trademark and the month and year of manufacture. The words "Oil and Gasoline Suction and Discharge 150 psi" (1034 kPa) shall be inlaid in the cover and vulcanized thereto. Hose 25 feet (7620 mm) or greater in length shall have brands inlaid in the cover at two places on each length beginning 4 feet (1219 mm) from the ends. The letters shall be not less than 0.250-inch (6 mm) high. Hose less than 25 feet (7620 mm) in length shall have one inlaid brand approximately in the center.

3.9 Workmanship. Hose and hose assemblies shall be manufactured and processed in such a manner as to be uniform in quality and shall be free from foreign material and other defects that will affect life, serviceability, strength, assembly or durability. Workmanship shall be such as to enable the hose and hose assemblies o meet the applicable performance requirements of this specification.

4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Conformance inspection (see 4.4).

4.2 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in accordance with the applicable test method referenced in the test procedures.

4.2.1 Test equipment and inspection facilities. Test and measuring equipment and inspection facilities of sufficient accuracy, quality, and quantity to permit performance of the required inspection shall be established and maintained by the contractor. The establishment and maintenance of a calibration system to control the accuracy of the measuring and test equipment shall be in accordance with ISO/IEC DIS 17025 and NCSL Z540.3 as applicable.

4.2.2 Responsibility for compliance. All items shall meet all requirements of sections 3 and 4. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

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4.2.3 Lot records. Manufacturers shall keep lot records for 3 years minimum. Manufacturers shall monitor for compliance to the prescribed procedures, and observe that satisfactory manufacturing conditions and records on lots are maintained for these hose assemblies. The records, including as a minimum, an attributes summary of all quality conformance inspections conducted on each lot, shall be available to review by customers at all times.

4.3 First article inspection. All samples shall be subjected to first article testing in accordance with table VI sequence is manufacturer's discretion.

4.3.1 Samples for first article. Samples for first article shall be representative of the products proposed to be furnished to this specification. Sampling for hose and hose assemblies shall be in accordance with 4.3.5.

4.3.2 First article inspection routine. First article inspection shall be performed at a laboratory acceptable to the procuring activity on sample units produced with equipment and procedures used in production.

4.3.3 Acceptance of first article inspection. A first article test cannot be waived by DLA unless the contractor has delivered the same item within the last three years, has no unfavorable quality history, and has not proposed changes to the processes or changed any subcontractors. DLA will not accept first article test results outside the stated requirements. All waivers or deviations shall be approved by the preparing activity; DLA Land and Maritime, Columbus, Attn: VAI, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to FluidFlow@dla.mil.

4.3.4 Failures. All samples must meet all of the contract requirements. Failure of a sample unit to pass any test shall be cause for rejection of the entire lot and to deny grant first article approval.

4.3.5 First article test samples. The first article test samples shall be in accordance with table VII.

4.3.6 First article production samples. Unless otherwise specified, after award of the contract or order, the manufacturer shall forward the required of hose or hose assemblies, fabricated from random samples of the hose or hose assemblies. The samples shall be representative of the construction, workmanship, components, and materials to be used during production. When a manufacturer is in continuous production of the hose and hose assemblies from one contract to another, submission of additional first article samples for a new contract may be waived at the discretion of the acquiring activity (see 6.2).

4.3.6.1 Disposition of samples. First article samples shall be furnished to the Government as directed by the contracting officer (see 6.2).

4.3.7 Waivers or deviations to specification requirements. All waivers or deviations to specification requirements shall be coordinated through the preparing activity: DLA Land and Maritime, Attn: VAI, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Fluidflow@dla.mil.

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TABLE VI. First article inspection.

Inspection	Requirement paragraph	Inspection paragraph
Visual	3.4.	4.6
Carcass	3.2.5	4.6
Hose body	3.4.1	4.6
Hose ends	3.4.4	4.6
Hand-built wire reinforced construction	3.4.4.1	4.6
Horizontal braided/spiraled wire reinforced construction	3.4.4.2	4.6
Textile reinforcement	3.4.5	4.6
Rubber filling strip	3.4.3	4.6
Nipples and flanges (steel bronze, or brass)	3.4.6	4.6
Integral flanged aluminum-alloy nipples	3.4.6.1	4.6
Integral swivel connection couplings	3.4.7	4.6
Marking	3.14	4.6
Length	3.3 and 3.3.1	4.7.1
Inside diameter	3.4	4.7.2
Tube and cover thicknesses	3.4	4.7.3
Resistance to vacuum	3.5	4.7.4
Hydrostatic pressure test	3.8.1	4.7.5
Stability (dimensional changes)	3.8.2	4.7.5
Electrical conductivity	3.10	4.7.19
Permanent elongation	3.8.2	4.7.5k
Electrical conductivity	3.10	4.7.19
Minimum bend test under pressure	3.8.3	4.7.6
Burst pressure test	3.8.4	4.7.7
Kink test	3.9	4.7.8
Initial tensile properties	3.11.1	4.7.9
Tensile properties after immersion	3.11.2	4.7.10
Volume change after immersion	3.6	4.7.11
Accelerated aging of compounds	3.11.3	4.7.12
Resistance to ozone	3.13	4.7.13
Initial adhesion	3.12	4.7.14
Adhesion after filling	3.12	4.7.15
Nonvolatile extractable material	3.7	4.7.16
Phosphate plasticizer test	3.4.2	4.7.17
Helix wire		4.7.18
Tensile strength	3.2.4	4.7.18.1
Flexibility	3.2.4.1	4.7.18.2

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TABLE VII. Samples.

Inspection	Samples	Inspection paragraph
Visual	All	4.6
Carcass	All	4.6
Hose body	All	4.6
Hose ends	All	4.6
Hand-built wire reinforced construction	All	4.6
Horizontal braided/spiraled wire reinforced construction	All	4.6
Textile reinforcement	All	4.6
Rubber filling strip	All	4.6
Nipples and flanges (steel, bronze, or brass)	All	4.6
Integral flanged aluminum-alloy nipples	All	4.6
Integral swivel connection couplings	All	4.6
Marking	All	4.6
Length	All	4.7.1
Inside diameter	All	4.7.2
Tube and cover thicknesses	End specimens 1/	4.7.3
Resistance to vacuum	4 foot hose	4.7.4
Hydrostatic pressure test	All hose assemblies	4.7.5
Stability (dimensional changes)		4.7.5
Electrical conductivity		4.7.19
Permanent elongation		4.7.5k
Electrical conductivity		4.7.19
Minimum bend test under pressure	Hose adequate to bend entire radius	4.7.6
Burst pressure test	8 foot hose assembly	4.7.7
Kink test	Hose adequate to bend entire radius	4.7.8
Initial tensile properties	Specimens 1/	4.7.9
Tensile properties after immersion	Specimens 1/	4.7.10
Volume change after immersion	Samples of the tube and cover 1/	4.7.11
Accelerated aging of compounds	Specimens 1/	4.7.12
Tensile strength 2/		4.7.9
Ultimate elongation 2/		4.7.9
Resistance to ozone	Sample of cover 1/	4.7.13
Initial adhesion	12 inch hose 3/	4.7.14
Adhesion after filling		4.7.15
Nonvolatile extractable material	Specimens 1/	4.7.16
Phosphate plasticizer test	Specimens 4/	4.7.17
Helix wire	6 feet 5/	4.7.18
Tensile strength		4.7.18.1
Flexibility		4.7.18.2

1/ Specimens from undamaged hose assembly from burst test.
2/ Compare results with initial tensile properties, see 4.7.9
3/ Hose shall be shall be without helix wire and nipples.
4/ Specimens of the residue remaining after a nonvolatile extractable material test in 4.7.16.
5/ Wire taken from same lot as hose and hose assemblies submitted for testing.

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4.4. Conformance inspection. For manufacturers that have successfully passed first article inspections and are continuously producing hose assemblies to this specification, on going inspections shall consist of individual inspections (see table VIII) and periodic inspections (see 4.5). If first article is waived due to prior successful first article inspection, the individual inspections and periodic inspections shall be the manufactures in house inspection procedures.

4.4.1. Individual inspection. Individual inspection shall consist of the inspections specified in table VIII in the order shown. Individual inspections shall be implemented on a continual basis throughout the production of hose assemblies.

TABLE VIII. Individual inspections.

Inspections	Requirement paragraph	Inspection paragraph	Number of samples
Visual inspection	3.4.	4.6	100%
Configuration (hose assembly only)	3.4.4	4.6	100%
Length	3.3 and 3.3.1	4.7.1	100%
Electrical conductivity	3.10	4.7.19	100%

4.4.2. Individual inspections. Individual inspection tests specified in table VIII shall be performed on a production lot basis.

4.4.2.1 Nonconformance individual inspections. If one or more defects are found in the inspection lot, then the production lot shall be screened for that particular defect and defects removed. An inspection lot shall be selected from the production lot and all individual tests (see table VIII) shall be performed. If one or more defects are found in the second inspection lot, the production lot shall be rejected and shall not be supplied to this specification. Test data of part performance shall be made available to the contracting agency upon request.

4.5 Periodic inspection. Periodic inspections shall consist of the inspections specified in table VI and shall be made on test samples which have been subjected to and passed the individual inspections (see table VIII).

4.5.4 Periodic inspection samples.

4.5.1 Lot. A lot shall consist of hose assemblies manufactured under essentially the same conditions and submitted for inspection at substantially the same time.

4.5.2 Lot samples. For each lot of five hundred hose assemblies or fraction thereof produced, one hose assembly sample shall be fabricated from random samples of the bulk hose and end fittings using production processes and procedures.

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4.5.3 Non conformance periodic inspections. In the event a failure should occur during periodic inspection tests, specified in table VI then the production lot shall be screened for that particular defect and defects removed. An inspection lot shall be selected from the production lot and all sampling and periodic tests shall be performed. If one or more defects are found in the second inspection lot, the production lot shall be rejected and shall not be supplied to this specification. Test data of part performance shall be made available to the contracting agency upon request.

4.6 Examination. Each of the sample hose lengths selected shall be visually and dimensionally examined to determine conformance with all the requirements of this specification not involving tests. The classification of defects in FED-STD-162 shall be used to determine and evaluate defects through visual inspection. In addition to the defects listed in FED-STD-162, the hose sample shall be examined for the following defect: Marking not as specified (see 3.14).

4.7 Tests.

4.7.1 Length. Hose selected shall be measured for length including fittings (see 3.3).

4.7.2 Inside diameter (see 3.4). Hose selected shall have the ID measured in accordance with ASTM D380 and shall meet the requirements of 3.4.

4.7.3 Tube and cover thicknesses (see 3.4). The thicknesses of the tube and cover shall meet the requirements of 3.4. The following details shall apply:

a. The thicknesses of the tube and cover shall be measured on specimens removed from each end of the hose section obtained in accordance with 4.2.
b. The thicknesses shall be determined by the procedures given in ASTM D380.
c. The respective values found for the tube thickness shall not be averaged, but shall be regarded as independent measurements, each of which shall meet the requirement.
d. The respective values for the cover thicknesses shall be treated in the same manner.

4.7.4 Resistance to vacuum (see 3.5). Hose when subjected to resistance to vacuum shall meet the requirements of 3.5. Hose selected shall be subjected for 5 minutes to a vacuum test equivalent to 20 inches (508 mm) of mercury. One end of the hose shall be equipped with a transparent cap and an electric light shall be used to permit visual examination for collapse or failure.

4.7.5 Hydrostatic pressure test (see 3.8.1). Hose assemblies when hydrostatically tested shall meet the requirements of 3.8.1). Each length of hose, complete with fittings, flanges, or adapters (or all), shall be subjected to a 250 psi (1724 kPa) pressure test as follows:

a. Lay out the hose as straight as possible.
b. Fill with water, venting to remove all air and apply a pressure of 10 psi (69 kPa).
c. Mark off a test length of 50 inches (1 270 mm), clear of the end reinforcement, for later measurement of elongation.
d. Increase the pressure, over a period of 5 minutes, from 10 psi (69 kPa) to half the test pressure of 250 psi (1724 kPa), (125 psi (862 kPa)), hold this for 10 minutes, then reduce pressure over a period of 5 minutes to zero.
e. Raise the pressure over a period of 5 minutes to the full test pressure of 250 psi (1724 kPa), and hold it for 10 minutes.
f. Before releasing the full test pressure, measure the distance between the test marks to ascertain the temporary elongation and record the increase as a percentage of the original length measured at 10 psi (69 kPa).

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g. While still under pressure perform the electrical conductivity test, see 4.7.19:
h. Reduce pressure over a period of 5 minutes to zero.
i. On the first hose assembly tested in each lot, after an interval of not less than 15 minutes, raise the pressure again to 10 psi (69 kPa). Measure the distance between the test marks to ascertain the permanent elongation and record the increase as a percentage of the original length measured at 10 psi (69 kPa).
j. No other hose in the lot will require testing provided that this hose meets the permanent elongation requirement see 3.8.2.
k. If the hose does not meet the permanent elongation requirement in accordance with 3.8.2, all hose in the lot shall be tested.
l. Release pressure and perform electrical conductivity test, see 4.7.19.

4.7.6 Minimum bend test under pressure (see 3.8.3). Hose selected shall be subjected to 150 psi (1034 kPa) hydrostatic pressure and bent to the appropriate radius specified in table II.

4.7.7 Burst pressure test (see 3.8.4). A hydrostatic pressure shall be applied equal to four times the working pressure of 150 psi (1034 kPa) (600 psi (4137 kPa)). This pressure shall be attained at a uniform rate of not less than 300 psi (2068 kPa) and not greater than 1,000 psi (6 895 kPa) per minute. The pressure shall then be raised until the hose fails; the pressure at which it fails shall be recorded and shall meet the requirements of 3.8.4.

4.7.8 Kink test (see 3.9). Empty hose when laid out on a flat surface in a curve drawn with a radius as given in table II shall attain this curved position without kinking or flattening.

4.7.9 Initial tensile properties (see 3.11.1). Initial tensile strength and ultimate elongation shall meet the requirement's of 3.11.1. The Initial tensile strength and ultimate elongation shall be determined by the procedure given in ASTM D412, using specimens cut with die C.

4.7.10 Tensile properties after immersion (see 3.11.2). The tensile strength and ultimate elongation after immersion in ASTM D471 reference fuel B for 70 ± 0.50 hours at 73 ± 4 °F (23 ± 2°C) shall be determined in accordance with ASTM D471, using specimens cut with die C of ASTM D412. Tensile strength shall be based on the swollen cross sectional area of the specimen.

4.7.11 Volume change after immersion (see 3.6). The change in volume of samples of the tube and cover after immersion in ASTM D471 reference fuel B for 70 ± 0.50 hours 73 ± 4°F (23 ± 2°C) shall be determined in accordance with ASTM D471. The change in volume shall be reported as a percent of the original volume and shall meet the requirements of 3.6.

4.7.12 Accelerated aging of compounds (see 3.11.3). The resistance of the tube and cover to accelerated aging shall meet the requirements of 3.11.3. The following details shall apply:

a. The resistance of the tube and cover to accelerated aging shall be determined in accordance with ASTM D573, using specimens cut with die C of ASTM D412, except that the time aging shall be 70 ± 0.50 hours at 212°F (100°C).
b. Tensile strength and ultimate elongation tests shall be used to determine the amount of deterioration.

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4.7.13 Resistance to ozone (see 3.13). The hose cover when subjected to ozone testing shall meet the requirements of 3.13. The following details shall apply:

a. The sample of cover stock shall be tested for ozone resistance and shall meet the requirements in 3.13.
b. Testing shall be in accordance with ASTM D1149 method B, procedure B2, except that after conditioning for 24 hours in an ozone-free atmosphere the looped sample shall be exposed for 72 hours at 104 ± 4°F (40 ± 2°C) to an atmosphere containing 50 parts per hundred million of ozone.

4.7.14 Initial adhesion (see 3.12). The adhesion of the tube and cover to the reinforcement, and of the plies of reinforcement to each other, shall be determined in accordance with the machine method of ASTM D413 and shall meet the requirements of 3.12.

4.7.14.1 Specimens. Strip or ring specimens shall be prepared in accordance with ASTM D380, except that the specimen shall be not less than of 8 inches (203 mm) in length. For determining adhesion between plies of reinforcement, the plies to be separated must be at opposing angles. Three specimens of each interface shall be tested.

4.7.15 Adhesion after filling (see 3.12). Hose when subjected to adhesion after filling shall meet the requirements of 3.12. The following details shall apply:

a. A 12-inch (305 mm) length of the hose shall be suitably stoppered and filled with aromatic hydrocarbon fluid conforming to ASTM D471 reference fuel B, and maintained at a temperature of +75°F, ± 5°F (+23°C, ± 3°C) for 48 hours.
b. Ring specimens, as described in ASTM D413, shall be cut from the 12-inch (305 mm) piece of hose and tested in accordance with ASTM D413 (machine method) within 15 minutes after the removal of the test fluid. Nonconformance to 3.12 shall constitute failure of this test.

4.7.16 Nonvolatile extractable material.

4.7.16.1 Test specimen. Each specimen shall consist of strips about 0.063-inch (2 mm) square by 2 inches (51 mm) long, totaling about 5 grams in weight, cut from buffed pieces of the tube of the hose.

4.7.16.2 Procedure. The following details shall apply:

a. The specimens shall be weighed and placed in an “extraction thimble in the “Extraction Apparatus with Block Tin Condenser” shown in ASTM D297.
b. One hundred milliliter (mL) of ASTM reference fuel A of ASTM D471 shall be added to the flask.
c. The specimen shall be extracted for 46 ± 0.50 hours at such a rate that 2.50 to 3.50 minutes are required to fill and empty the extraction cup. The extract shall be poured into a weighed container.
d, The specimen and flask shall be rinsed with 25 mL of fresh fuel A and the wash added to the extract.
e. The nonvolatile residue shall be determined from the extract according to ASTM D381, except that the evaporation time shall be 45 minutes.
f. The weight of the residue shall be expressed as percent of the original weight of the specimen. The test shall be performed in duplicate and results averaged.

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4.7.17 Phosphate plasticizer test (see 3.4.2). When testing for phosphate plasticizer hose shall meet the requirements of 3.4.2. The following details shall apply:

a. Phosphate plasticizer shall be determined qualitatively in accordance with method 15825 of FED-STD-601.
b. Use a small piece of clean sodium about the size of a pea, a test tube about 3 inches (76 mm) long, and the specimens of the residue remaining after a nonvolatile extractable material test as specified in 4.7.16.
c. The procedure shall be repeated if phosphorus is found, omitting addition of the nonvolatile extractable material in order to determine whether phosphorus was present in any form in any of the reagents used.

4.7.17.1 Phosphate plasticizer test, alternate method. An alternate method for determining phosphate plasticizer qualitatively shall be in accordance with ASTM D2702. The following details shall apply:

a. Use the specimen of residue remaining after the nonvolatile extractable material test as specified in 4.7.16.
b. The procedure shall be repeated if phosphorus is found, omitting the addition of the nonvolatile extractable material in order to determine whether phosphorus was present in any form in any of the reagents used.
c. Hose shall meet the requirements of 3.4.2.

4.7.18 Helix wire.

4.7.18.1 Tensile strength (see 3.2.4). The tensile strength of the wire shall be determined in accordance with ASTM E8/E8M and shall meet the requirements of 3.2.4.

4.7.18.2 Flexibility (see 3.2.4.1). The helix wire or wires shall be wrapped six full tight turns around a mandrel of a diameter three times the diameter of the wire. The wrapped wire shall show no defects in the steel after wrapping and meet the requirements of 3.2.4.1. Flaking or peeling of wire coating shall not be considered a defect.

4.7.19 Electrical conductivity (see 3.10). While subjected to 250 psi (1724 kPa) pressure during the hydrostatic pressure test (see 4.7.5), each length of hose shall be tested for electrical conductivity in an electrical circuit. Any hose assembly that does not meet electrical conductivity requirements see 3.10, shall be rejected. The following details shall apply:

a. While the hose assembly is pressurized, see 4.7.5g apply the voltmeter leads to the metal end fittings.
b. Apply a test voltage (9V) and measure the resistance 5 seconds ± 1 second after the voltage is applied.
c. Resistance shall be in accordance with 3.10.
d. After pressure has been released during the hydrostatic pressure test in 4.7.5k apply a test voltage (9V) and measure the resistance 5 seconds ± 1 second after the voltage is applied.
e. Resistance shall be in accordance with 3.10.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which is helpful, but is not mandatory.)

6.1 Intended use. Hose and hose assemblies are intended for use in the transfer of oil and gasoline from anchored tankers and barges to shore stations capable of handling discharge pressures up to 150 psi (1034 kilopascal (kPa)).

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. PIN see 1.2.
c. Quantity required.
d. Whether first article inspection is waived (see 4.3.3).
e. Lot records if required (4.2.3).
f. Name and address of the first article inspection test facility to which first article test samples (if required) are to be forwarded (see 4.3) and the name and address of the Government activity responsible for conducting the first article inspection program (see 4.3. and 6.3).
g. Name and address of the facility to have first article production sample(s) sent, see 4.3.6. specify quantity and type of samples.
h. Packaging requirements (see 5.1).

6.3 First article. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first article samples. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 End fittings. The 4-inch integral swivel couplings as specified in 3.4.4.1 and 3.4.7 may be obtained from Moon-Allenco, Easton, PA, with the designation "2950 Integral Nipples." End fittings specified in 3.4.4.2 may be obtained from Mulconroy Division, Dixon Valve and Coupling Company, Philadelphia, PA, with the designation Mulconroy style TM "Holedall" coupling.

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6.5 Hazardous substances and ozone depleting chemicals. The ozone resistance test (see 4.7.13) may contain hazardous chemicals. It should be handled in accordance with Federal regulations and guidelines to perform those tests. For further information about toxic chemicals and hazardous materials list, consult the Environmental Protection Agency web database at www.epa.gov/ebtpages/pollutants.html.

6.6 National Stock Numbers (NSNs). The following is a list of NSNs assigned which correspond to this federal specification. The list may not be indicative of all possible NSNs associated with the federal specification.

NSN	Nomenclature
4720-00-034-7813	Hose Assembly, Nonmetallic.
4720-00-200-0351	Hose Assembly, Nonmetallic.
4720-00-200-0353	Hose Assembly, Nonmetallic.
4720-00-230-6516	Hose Assembly, Nonmetallic.
4720-00-289-1408	Hose Assembly, Nonmetallic.

6.7 Subject term (key word) listing.

Conductive
Fuels
Ozone
Petroleum transfer

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodian:
Navy - YD
DLA - CC

Preparing Activity:
DLA - CC

(Project 4720-2018-011)

Review Activity:
Navy - SH

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